



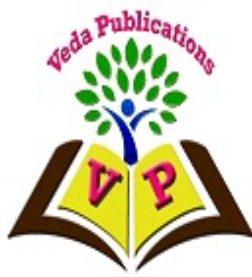
RESEARCH ARTICLE



NUMERALS IN KAIPENG AND SIMTE

Lorina D Tariang ^{1*}, H. Kapginlian²¹(Research Scholar, Assam University, Silchar)²(Research Scholar, North Eastern Hill University, Shillong)Doi: <https://doi.org/10.54513/JOELL.2024.12202>

ABSTRACT



Kaipeng and Simte belong to the Kuki-Chin sub-group of the Tibeto-Burman language family. Kaipeng belongs to the Old Kuki sub-group while Simte belongs to the Northern Kuki-Chin group (Grierson 1903). Kaipeng is spoken in Tripura by approximately 15000 speakers. According to 2011 Census, Simte constitutes a population of 6728 (Census Report, 2011). This paper attempts to present a comparative analysis of the numeral system in two Kuki-Chin languages, precisely Kaipeng and Simte. The numeral systems are described, compared and analysed in the realm of cardinals, ordinals, fractional, multiplicative, distributive and approximate numerals.

Article history:

Received on : 04-04-2025

Accepted on : 28-04-2025

Available online: 13-05-2025

Keywords: *Approximate, Distributive, Fractional, Kaipeng, Simte.*



INTRODUCTION

Kaipeng is a language spoken in the North-Eastern state of Tripura and are concentrated in four districts namely: Khowai, Gomati, North Tripura and Sepahijala. The total number of Kaipeng speakers in Tripura is approximately 15000.¹ The term Kaipeng is used to represent both the language and the people. The word '*Kaipeng*' is derived from two words *Kai* meaning 'to lead' and *peng* meaning 'another way or path'. Thus, Kaipeng means leading someone towards another path. Linguistically, Kaipeng is not mentioned in any of the classification, but Kaipeng is a sub-group of Halam and Halam is classified under Old Kuki along with other languages like Kom, Bete, Aimol, Langrong, Anal, Aimol and Chothe (Thurgood 2003). Simte is also one of the Kuki-Chin language family inhabiting in the North-Eastern part of India. They are predominantly concentrated in Churachandpur and Pherzawl districts of Manipur (Hangluah, 2019). According to Suantak and Singh (2011), scores of Simte population are scattered in Sikkim, Arunachal Pradesh, Nagaland, Mizoram, Meghalaya, Tripura and Assam. The word '*Simte*' is a combination of two morphemes: '*sim*' which refers to the southern region and '*te*' which meaning 'people'. This language is also classified under the Northern Kuki-Chin groups along with its genetic language family such as Paite, Zou, Vaiphei, Gangte and Thadou (Thurgood, 2003). It is observed that both the languages exhibit pronoun dropping and SOV word order.

1. LITERATURE REVIEW

Hammarstrom (2009) posits that numerals are normed spoken expressions used to denote the

precise quantity of objects within a particular social context, reflecting the collective understanding of the speech community. Numerals represent a critical aspect of morphosyntactic studies across languages, given their foundational role in language structure. This is especially significant in the context of younger generations, who increasingly favour English numerals over traditional indigenous counting systems. The widespread use of English numerals can be attributed to the pervasive influence of Western education, which often overlooks indigenous counting systems in the curriculum. As a result, this shift leads to the frequent occurrence of code-switching and code-mixing, particularly in the realm of numeration. In agreement with this perspective, Saikia and Mamta (2020) highlights that numeral systems are especially vulnerable to extinction, particularly in minority and lesser-documented languages. This phenomenon is observable in the case of both the Kaipeng and Simte languages. Pioneer studies on numerals in Kuki-Chin are conducted by Grierson in 1904. The studies include languages such as Paite, Lushai and Thadou. However, these studies do not provide an exhaustive analysis of numerals in both Kaipeng and Simte languages.

In the context of Kaipeng, it is noteworthy that a thorough review of existing literature revealed a significant gap in knowledge regarding the numeral system employed in Kaipeng. Specifically, no prior studies had been conducted to investigate the intricacies of their numerical system, leaving a void in understanding this fundamental aspect of their language.



Previous studies on Simte by Singh (2009) and Champeon (2019) discuss the cardinal and ordinal numerals. However, they failed to describe the distributive, fractional and approximate numerals. Additionally, no comparative linguistic research has been conducted on both Kaipeng and Simte languages. This paper represents a pioneering effort in the comparative analysis of a linguistic aspect between the two languages. It seeks to fill this gap by providing a detailed description and analysis of the numeral systems in both Kaipeng and Simte.

2. NUMERAL SYSTEM IN KAIPENG AND SIMTE

Numeral system in Kaipeng and Simte are of the decimal counting system. The numerals in these languages can be classified into cardinal, ordinal, fractional, multiplicative, distributive and

approximate. Like most of the other Kuki-Chin languages, higher numbers are formed by joining the basic numbers through the process of addition and multiplication in these languages. Suffixes and prefixes are added to the basic numerals to form different types of numerals.

3. CARDINAL NUMBERS

Cardinal Numerals in the two languages can be divided into two types: (i) Basic Cardinal numbers and (ii) Compound Cardinal numbers.

3.1. BASIC CARDINAL NUMBERS

The numbers from one to ten are considered as the basic form. In Kaipeng, *-ka* is suffixed to the numbers. In Simte, on the other hand, all the basic numbers have distinct lexical items and follows monosyllabic pattern with an exception for *sagi?* 'seven'.

TABLE 1. BASIC CARDINAL NUMERALS IN KAIPENG AND SIMTE

Kaipeng	Simte	Gloss
<i>k^hat -ka</i>	<i>k^hat</i>	'one'
<i>ni -ka</i>	<i>ni?</i>	'two'
<i>t^hum -ka</i>	<i>t^hum</i>	'three'
<i>li -ka</i>	<i>li</i>	'four'
<i>rəŋa -ka</i>	<i>ŋa</i>	'five'
<i>ru -ka</i>	<i>gup</i>	'six'
<i>səri -ka</i>	<i>sagi?</i>	'seven'
<i>riat -ka</i>	<i>giat</i>	'eight'
<i>kua -ka</i>	<i>kua</i>	'nine'
<i>som -ka</i>	<i>som</i>	'ten'



3.2. COMPOUND CARDINAL NUMBERS

Numerals in both the languages are also formed by the process of compounding. Numerals are classified into different categories based on how they are formed. In Kaipeng and Simte compound numerals can be classified into three categories: (i) additive compound numerals, (ii) multiplicative compound numerals and (iii) multiplicative-cum-additive compound numerals.

3.2.1. ADDITIVE COMPOUND NUMERALS

Numbers from eleven to nineteen are additive compound numerals in the languages. The numbers are formed by combining ten to the basic numbers from one to nine. In Kaipeng the numbers are combined with the conjunctive participle *lei* and in Simte it is combined with *le?*. However, the pattern is the same for the two languages. For example: $10+1=11$, $10+2=12$, $10+3=13$ etc. The additive compound numerals are shown in Table 2

TABLE 2. ADDITIVE COMPOUND NUMERALS IN KAIPENG AND SIMTE

Kaipeng	Simte	Gloss
<i>sɔm lei k^hatka</i> [10+1=11]	<i>sɔm le? k^hat</i> [10+1=11]	'eleven'
<i>sɔm lei nika</i> [10+2=12]	<i>sɔm le? ni?</i> [10+2=12]	'twelve'
<i>sɔm lei t^humka</i> [10+3=13]	<i>sɔm le? t^hum</i> [10+3=13]	'thirteen'
<i>sɔm lei lika</i> [10+4=14]	<i>sɔm le? li</i> [10+4=14]	'fourteen'
<i>sɔm lei rəŋaka</i> [10+5=15]	<i>sɔm le? ŋa</i> [10+5=15]	'fifteen'
<i>sɔm lei ruka</i> [10+6=16]	<i>sɔm le? gup</i> [10+6=16]	'sixteen'
<i>sɔm lei sərɪka</i> [10+7=17]	<i>sɔm le? sagi?</i> [10+7=17]	'seventeen'
<i>sɔm lei riatka</i> [10+8=18]	<i>sɔm le? giat</i> [10+8=18]	'eighteen'
<i>sɔm lei kuaka</i> [10+9=19]	<i>sɔm le? kua</i> [10+9=19]	'nineteen'

**3.2.2. MULTIPLICATIVE COMPOUND****NUMERALS**

Large numbers are formed by the process of multiplication in the two languages. The multiplicative compound numerals can be further divided into (i) lower multiplicative compound and (ii) higher multiplicative compound.

3.2.2.1. LOWER MULTIPLICATIVE COMPOUND

The numbers from ten to ninety are formed by multiplying the number ten to the basic numbers from one to nine. For example: $10 \times 1 = 10$, $10 \times 2 = 20$, $10 \times 3 = 30$ etc.

TABLE 3. LOWER MULTIPLICATIVE COMPOUND NUMERALS IN KAIPENG AND SIMTE

Kaipeng	Simte	Gloss
<i>sɔm ka</i> [10X1=10]	<i>sɔm</i> [10X1=10]	'ten'
<i>sɔm nika</i> [10X2=20]	<i>sɔm niʔ</i> [10X2=20]	'twenty'
<i>sɔm thumka</i> [10X3=30]	<i>sɔm tʰum</i> [10X3=30]	'thirty'
<i>sɔm lika</i> [10X4=40]	<i>sɔm li</i> [10X4=40]	'fourty'
<i>sɔm rəŋaka</i> [10X5=50]	<i>sɔm ŋa</i> [10X5=50]	'fifty'
<i>sɔm ruka</i> [10X6=60]	<i>sɔm gup</i> [10X6=60]	'sixty'
<i>sɔm sərika</i> [10X7=70]	<i>sɔm sagiʔ</i> [10X7=70]	'seventy'
<i>sɔm riatka</i> [10X8=80]	<i>sɔm giat</i> [10X8=80]	'eighty'
<i>sɔm kuaka</i> [10X9=90]	<i>sɔm kua</i> [10X9=90]	'ninety'

3.2.2.2 HIGHER MULTIPLICATIVE COMPOUND

The numbers from one hundred to nine hundred are also formed by the process of multiplication. Hundred is multiplied with the number one to nine. For instance, $100 \times 1 = 100$, $100 \times 2 = 200$, $100 \times 3 = 300$.

**TABLE 4. HIGHER MULTIPLICATIVE COMPOUND NUMERALS IN KAIPENG AND SIMTE**

Kaipeng	Simte	Gloss
rəza ka [100X1=100]	za k ^h at [100X1=100]	'one hundred'
rəza nika [100X2=200]	za niʔ [100X2=200]	'two hundred'
rəza t ^h umka [100X3=300]	za t ^h um [100X3=300]	'three hundred'
rəza lika [100X4=400]	za li [100X4=400]	'four hundred'
rəza rəŋaka [100X5=500]	za ŋa [100X5=500]	'five hundred'
rəza ruka [100X6=600]	za gup [100X6=600]	'six hundred'
rəza sərɪka [100X7=700]	za saɣiʔ [100X7=700]	'seven hundred'
rəza riatka [100X8=800]	za ɣiat [100X8=800]	'eight hundred'
rəza kuaka [100X9=900]	za kua [100X9=900]	'nine hundred'

The numbers one thousand to nine thousand are similarly formed by multiplying thousand to the basic numbers one to nine, that is 1000X1=1000, 1000X2=2000, 1000X3=3000 etc.

TABLE 5. HIGHER MULTIPLICATIVE COMPOUND NUMERALS IN KAIPENG AND SIMTE

Kaipeng	Simte	Gloss
saŋ ka [1000X1=1000]	saŋ k ^h at [1000X1=1000]	'one thousand'
saŋ nika [1000X2=2000]	saŋ niʔ [1000X2=2000]	'two thousand'
saŋ t ^h umka [1000X3=3000]	saŋ t ^h um [1000X3=3000]	'three thousand'



<i>saŋ lika</i> [1000X4=4000]	<i>saŋ li</i> [1000X4=4000]	'four thousand'
<i>saŋ rəŋaka</i> [1000X5=5000]	<i>saŋ ŋa</i> [1000X5=5000]	'five thousand'
<i>saŋ ruka</i> [1000X6=6000]	<i>saŋ gup</i> [100X6=600]	'six thousand'
<i>saŋ sərika</i> [1000X7=7000]	<i>saŋ sagi?</i> [100X7=700]	'seven thousand'
<i>saŋ riātka</i> [1000X8=8000]	<i>saŋ giat</i> [100X8=800]	'eight thousand'
<i>saŋ kuaka</i> [1000X9=9000]	<i>saŋ kua</i> [100X9=900]	'nine thousand'

3.2.3 MULTIPLICATIVE-CUM-ADDITIVE COMPOUND NUMERALS

Numerals like 21-29, 31-39, 101-109, 201-209, 1001-1009 etc are formed by multiplicative cum additive method. In the case of Kaipeng, a base-10 multiplier is applied sequentially to the first three numbers, with the result added to the fourth number. For example, $10 \times 2 \times 1 + 2 = 22$, $10 \times 3 \times 1 + 2 = 32$. In contrast, Simte's system involves multiplying the base-10 only by the first number, then adding the third number, producing identical results: $10 \times 2 + 2 = 22$, $10 \times 3 + 2 = 32$. In Kaipeng, the conjunctive participle is dropped while in Simte the conjunctive participle 'le?' remains constant. The Multiplicative cum additive Compound Numerals in Kaipeng and Simte is shown in Table 6.

TABLE 6. MULTIPLICATIVE-CUM-ADDITIVE COMPOUND NUMERALS IN KAIPENG AND SIMTE

Kaipeng	Simte	Gloss
<i>səm t^hum k^hatnika</i>	<i>səmni? le? ni?</i>	'thirty two'
<i>səm lik k^hat t^humka</i>	<i>səmli le? t^hum</i>	'forty three'
<i>səm rəŋak k^hatnika</i>	<i>səm ŋa le? ni?</i>	'fifty two'
<i>səm sərik k^hatnika</i>	<i>səm sagi? le? ni?</i>	'seventy two'
<i>rəzat^hum k^hatt^humka</i>	<i>zat^hum le? t^hum</i>	'three hundred three'
<i>rəzalik k^hatlika</i>	<i>zali le? li</i>	'four hundred four'
<i>saŋrəŋak arəŋakka</i>	<i>saŋŋa le? ŋa</i>	'five thousand five'



<i>saŋruk arukka</i>	<i>sangup leʔ gup</i>	'six thousand six'
----------------------	-----------------------	--------------------

4. ORDINAL NUMERALS

Ordinal numbers are formed by the process of prefixation and suffixation in the languages. In Kaipeng /a/ is prefixed and /na/ is suffixed to form ordinal numbers. However, the prefix /a/ is absent in

the formation of first in Kaipeng, where the prefix is dropped but is followed by the suffix /na/. In Simte, on the other hand, to form ordinal numbers /na/ is suffixed to the basic numbers. Table 7 provides the ordinal numbers in the languages.

TABLE 7. ORDINAL NUMERALS IN KAIPENG AND SIMTE

Kaipeng	Simte	Gloss
<i>k^hatka-na</i>	<i>k^hat-na</i>	'first'
<i>a-nika-na</i>	<i>niʔ-na</i>	'second'
<i>a-t^humka-na</i>	<i>t^hum-na</i>	'third'

5. FRACTIONAL NUMERALS

In Kaipeng and Simte, fractional values can be expressed using phrases such as half, one and half, two-thirds, one-tenth, and so on. In Kaipeng, the morphological marker for the concept of 'half' is represented by the morpheme 'bɔŋ' while its Simte counterpart is realized by 'kim'. which is added as a

prefix to the numeral 'khat', signifying the number 'one'. In the fraction of 'one and half', the morpheme 'khat' is succeeded by 'bɔŋ'. Interestingly, both the morphemes are suffixed by 'ka'. Whereas in Simte both the one and half are joined by the conjunctive particle 'leʔ'. In numeral containing 'one-tenth' the morpheme 'a' connects both the number ten and one in both the languages.

TABLE 8. FRACTIONAL NUMERALS IN KAIPENG AND SIMTE

Kaipeng	Simte	Gloss
<i>bɔŋka</i>	<i>kim k^hat</i>	'half'
<i>k^hat -ka bɔŋ -ka</i>	<i>k^hat leʔ kim</i>	'one and half'
<i>sɔmk^hat -a -k^hatka</i>	<i>sɔm-a-k^hat</i>	'one-tenth'

6. MULTIPLICATIVE NUMERALS

Multiplicative numerals in Kaipeng are formed by prefixing *vei* to the basic numbers. On the other hand, in Simte it is formed by suffixing *vei* to the basic numbers.

**TABLE 9. MULTIPLICATIVE NUMERALS IN KAIPENG AND SIMTE**

Kaipeng	Simte	Gloss
<i>vei-ka</i>	<i>k^hat -vei</i>	'once'
<i>vei-nika</i>	<i>ni? -vei</i>	'twice'
<i>vei-t^humka</i>	<i>t^hum -vei</i>	'thrice'

7. DISTRIBUTIVE NUMERALS

Distributive numerals in Kaipeng are formed by partial reduplication. But the distributive numerals in Simte is realized by the morpheme *chiat*. Table 10 provides the distributive numerals.

TABLE 10. DISTRIBUTIVE NUMERALS IN KAIPENG AND SIMTE

Kaipeng	Simte	Gloss
<i>k^hat k^hatka</i>	<i>k^hat c^hiat</i>	'one each'
<i>anik nika</i>	<i>ni? c^hiat</i>	'two each'
<i>at^hum t^humka</i>	<i>t^hum c^hiat</i>	'three each'

8. APPROXIMATE NUMERALS

Approximate numerals in Kaipeng are formed by suffixing /*tsaŋ*/ to the cardinal numbers. In Simte, on the other hand, the approximative numerals are formed by adding the suffix '-*vel*' to the cardinal numerals. Table 11 provides the approximate numerals of the languages.

TABLE 11. APPROXIMATE NUMERALS IN KAIPENG AND SIMTE

Kaipeng	Simte	Gloss
<i>sɔmka tsaŋ</i>	<i>sɔm -vel</i>	'about ten'
<i>rəzaka tsaŋ</i>	<i>za k^hat -vel</i>	'about hundred'
<i>saŋnikka tsaŋ</i>	<i>za ni? -vel</i>	'about two thousand'

9. CONCLUSION

A comparative analysis of the numeral systems in Kaipeng and Simte reveals both notable similarities and differences. In terms of basic cardinals, Kaipeng attaches the suffix *-ka* to the basic numeral, whereas Simte features an idiosyncratic cardinal numeral.

Both languages use compound numerals, including additive, multiplicative, and multiplicative cum additive compound numerals, to form higher cardinal numbers. The conjunctive particle *lei* is used in Kaipeng to link higher compound numerals, while



Simte employs *le?* for the same purpose. Additionally, both languages employ the suffix *-na* attached to the base numeral to form ordinal numerals. A significant commonality is the use of *vei* to indicate multiplicative numerals. However, a syntactic difference is observed: in Kaipeng, *vei* is prefixed to the base numeral, while in Simte, it is suffixed.

REFERENCES

- Champeon, C. (2019). *Simte Writers Handbook*. U.S.A: Bibles International.
- Directorate of Census Operation (2011), *Census of India*, Series 15 Manipur Part VIII (II), Director of Census operation.
- Grierson, G.A. (1904). *Linguistic Survey of India. Vol.3, Part III*. Delhi: Motilal Banarasidas.
- Hammarstorm, H. (2009). *Unsupervised Learning of Morphology and the Languages of the world*. PhD Thesis, University of Gothenburg.
- Hangluah, T.K.L.(2019). The Simte . In STC (Ed.), *60 Diamond Jubilee Simte Tribe Council Souvenir* (pp.16-17). Churachandpur: STC Publication.
- Saikia, A., Mamta, K. (2020) *A comparative analysis of Kuki-Chin numeral system*.
- Subramanian, V.M. (2020). (Ed.)Working Papers on Linguistic and Literature, 90-97. Coimbatore: Bharathiar University.
- Singh, N. B. (2009) . *Simte Grammar*. [Ph.D. Thesis, Department of Linguistics]. Imphal: Manipur University.
- Singh, S.K., & Suantak, K.S.K. (2011). *Epilogue. In Linguistics Ecology: Manipur*, (pp.285- 292). Guwahati: EBH Publishers (India).
- Thurgood, G., & Lapolla, R. J. (2003). *The Sino-Tibetan Languages*. London: Routledge.